

Лабораторная Работа №15. Динамическая маршрутизация

Администрирование локальных сетей

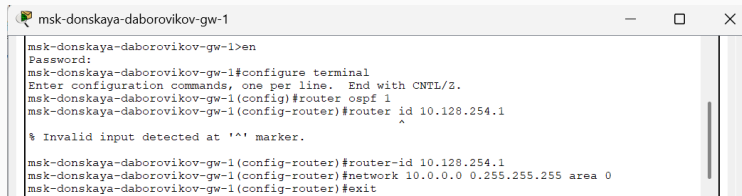
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Настроить динамическую маршрутизацию между территориями организации.

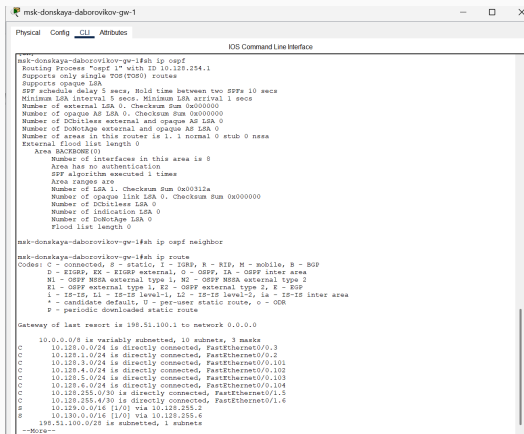
Настройка OSPF

A terminal window titled 'msk-donskaya-daborovikov-gw-1' with standard window controls. The terminal shows a sequence of commands: 'en' to enter enable mode, 'configure terminal' to enter configuration mode, 'router ospf 1' to start OSPF configuration, 'router id 10.128.254.1' to set the router ID (with an '^' character indicating a carriage return), and 'network 10.0.0.0 0.255.255.255 area 0' to define the OSPF network. An error message '% Invalid input detected at '^' marker.' appears after the router ID command. Finally, 'exit' is used to leave the configuration mode.

```
msk-donskaya-daborovikov-gw-1>en
Password:
msk-donskaya-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#router ospf 1
msk-donskaya-daborovikov-gw-1(config-router)#router id 10.128.254.1
^
% Invalid input detected at '^' marker.
msk-donskaya-daborovikov-gw-1(config-router)#router-id 10.128.254.1
msk-donskaya-daborovikov-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-donskaya-daborovikov-gw-1(config-router)#exit
```

Figure 1: Настройка OSPF на маршрутизаторе msk-donskaya-daborovikov-gw-1

Проверка состояния OSPF



```
msk-donskaya-daborovikov-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

msk-donskaya-daborovikov-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs, Minimum LSA arrival 1 secs
Number of external LSA 0, Checksum Sum 0x000000
Number of opaque AS LSA 0, Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
Area BACKBONE (0)
Number of interfaces in this area is 8
Area has no authentication
SPF algorithm executed 1 times
Area ranges are
Number of LSA 1, Checksum Sum 0x00312a
Number of opaque link LSA 0, Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

msk-donskaya-daborovikov-gw-1#sh ip ospf neighbor

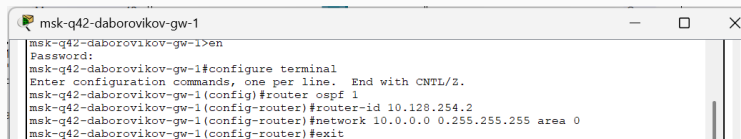
msk-donskaya-daborovikov-gw-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
p - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
C 10.128.0.0/24 is directly connected, FastEthernet0/0.3
C 10.128.1.0/24 is directly connected, FastEthernet0/0.2
C 10.128.3.0/24 is directly connected, FastEthernet0/0.101
C 10.128.4.0/24 is directly connected, FastEthernet0/0.102
C 10.128.5.0/24 is directly connected, FastEthernet0/0.103
C 10.128.6.0/24 is directly connected, FastEthernet0/0.104
C 10.128.255.0/30 is directly connected, FastEthernet0/1.5
C 10.128.255.4/30 is directly connected, FastEthernet0/1.6
S 10.129.0.0/16 [1/0] via 10.128.255.2
S 10.130.0.0/16 [1/0] via 10.128.255.6
198.51.100.0/28 is subnetted, 1 subnets
--More--
```

Figure 2: Проверка состояния протокола OSPF на маршрутизаторе mskdonskaya-daborovikov-gw-1 (просмотр статуса всех соседей в заданном сегменте, вывод информации из таблицы маршрутизации).

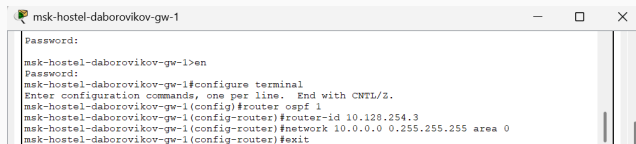
Настройка маршрутизатора



```
msk-q42-daborovikov-gw-1
msk-q42-daborovikov-gw-1>en
Password:
msk-q42-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-daborovikov-gw-1(config)#router ospf 1
msk-q42-daborovikov-gw-1(config-router)#router-id 10.128.254.2
msk-q42-daborovikov-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-q42-daborovikov-gw-1(config-router)#exit
```

Figure 3: Настройка маршрутизатора msk-q42-daborovikov-gw-1.

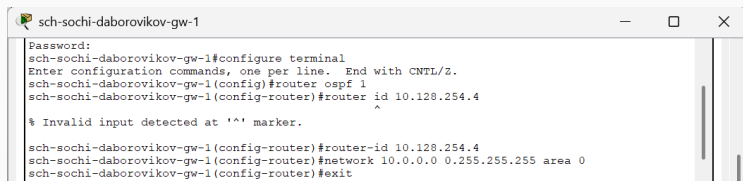
Настройка маршрутизирующего коммутатора



```
msk-hostel-daborovikov-gw-1
Password:
msk-hostel-daborovikov-gw-1>en
Password:
msk-hostel-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-daborovikov-gw-1(config)#router ospf 1
msk-hostel-daborovikov-gw-1(config-router)#router-id 10.128.254.3
msk-hostel-daborovikov-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-hostel-daborovikov-gw-1(config-router)#exit
```

Figure 4: Настройка маршрутизирующего коммутатора msk-hostel-daborovikov-gw1.

Настройка маршрутизатора

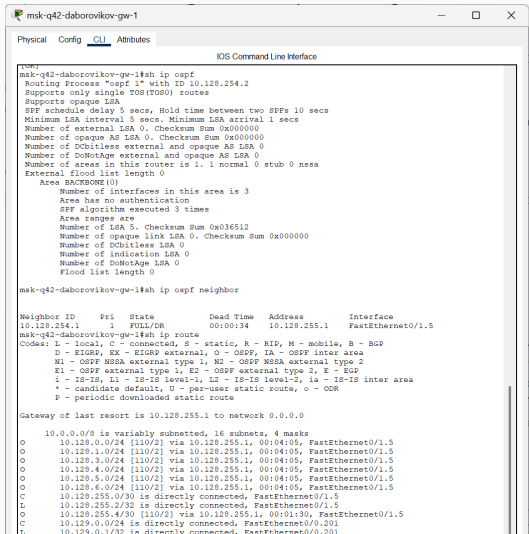


```
sch-sochi-daborovikov-gw-1
Password:
sch-sochi-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-daborovikov-gw-1(config)#router ospf 1
sch-sochi-daborovikov-gw-1(config-router)#router id 10.128.254.4
                                     ^
% Invalid input detected at '^' marker.

sch-sochi-daborovikov-gw-1(config-router)#router-id 10.128.254.4
sch-sochi-daborovikov-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
sch-sochi-daborovikov-gw-1(config-router)#exit
```

Figure 5: Настройка маршрутизатора sch-sochi-daborovikov-gw-1.

Проверка состояния OSPF



```
msk-q42-daborovikov-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

msk-q42-daborovikov-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.2
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs, Minimum LSA arrival 1 secs
Number of external LSA 0, Checksum Sum 0x0000000
Number of opaque AS LSA 0, Checksum Sum 0x0000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 3
    Area has no authentication
    SPF algorithm executed 3 times
    Area ranges are
    Number of LSA 5, Checksum Sum 0x036512
    Number of opaque link LSA 0, Checksum Sum 0x0000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

msk-q42-daborovikov-gw-1#sh ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.128.254.1      1    FULL/DR         00:00:34    10.128.255.1   FastEthernet0/1.5

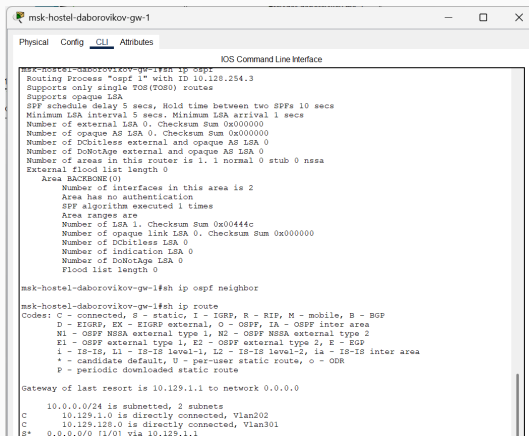
msk-q42-daborovikov-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 16 subnets, 4 masks
O       10.128.0.0/24 [110/2] via 10.128.255.1, 00:04:05, FastEthernet0/1.5
O       10.128.1.0/24 [110/2] via 10.128.255.1, 00:04:05, FastEthernet0/1.5
O       10.128.3.0/24 [110/2] via 10.128.255.1, 00:04:05, FastEthernet0/1.5
O       10.128.4.0/24 [110/2] via 10.128.255.1, 00:04:05, FastEthernet0/1.5
O       10.128.5.0/24 [110/2] via 10.128.255.1, 00:04:05, FastEthernet0/1.5
O       10.128.6.0/24 [110/2] via 10.128.255.1, 00:04:05, FastEthernet0/1.5
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.2/32 is directly connected, FastEthernet0/1.5
O       10.128.255.4/30 [110/2] via 10.128.255.1, 00:01:30, FastEthernet0/1.5
C       10.129.0.0/24 is directly connected, FastEthernet0/0.201
L       10.129.0.1/32 is directly connected, FastEthernet0/0.201
```

Figure 6: Проверка состояния протокола OSPF на маршрутизаторе

Проверка состояния OSPF



```
msk-hostel-daborovikov-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

msk-hostel-daborovikov-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.3
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs, Minimum LSA arrival 1 secs
Number of external LSA 0, Checksum Sum 0x000000
Number of opaque AS LSA 0, Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1, 1 normal 0 stub 0 nssa
External flood list length 0
Area BACKBONE(0)
  Number of interfaces in this area is 2
  Area has no authentication
  SPF algorithm executed 1 times
  Area ranges are
  Number of LSA 1, Checksum Sum 0x00444c
  Number of opaque link LSA 0, Checksum Sum 0x000000
  Number of DCbitless LSA 0
  Number of indication LSA 0
  Number of DoNotAge LSA 0
  Flood list length 0

msk-hostel-daborovikov-gw-1#sh ip ospf neighbor

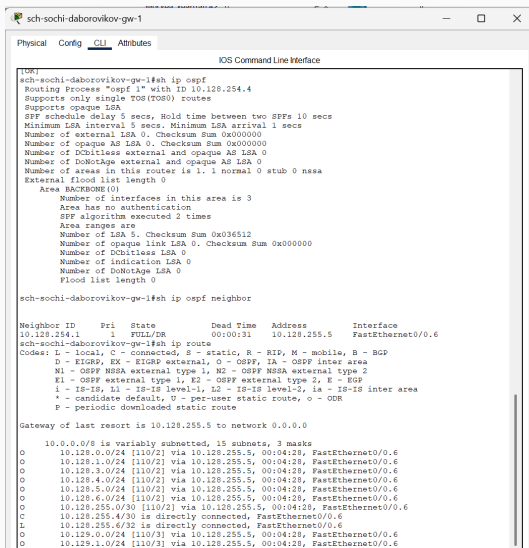
msk-hostel-daborovikov-gw-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        I1 - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 10.129.1.1 to network 0.0.0.0

    10.0.0.0/24 is subnetted, 2 subnets
C       10.129.1.0 is directly connected, Vlan202
C       10.129.128.0 is directly connected, Vlan301
S*    0.0.0.0/0 [1/0] via 10.129.1.1
```

Figure 7: Проверка состояния протокола OSPF на маршрутизирующем коммутаторе msk-hostel-daborovikov-gw-1.

Проверка состояния OSPF



```
sch-sochi-daborovikov-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

[OK]
sch-sochi-daborovikov-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.4
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs, Minimum LSA arrival 1 secs
Number of external LSA 0, Checksum Sum 0x000000
Number of opaque AS LSA 0, Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1, 1 normal 0 stub 0 nssa
External flood list length 0
Area BACKBONE(0)
Number of interfaces in this area is 3
Area has no authentication
SPF algorithm executed 2 times
Area ranges are
Number of LSA 5, Checksum Sum 0x036512
Number of opaque link LSA 0, Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

sch-sochi-daborovikov-gw-1#sh ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
10.128.254.1 1 FULL/DR 00:00:31 10.128.255.5 FastEthernet0/0.6

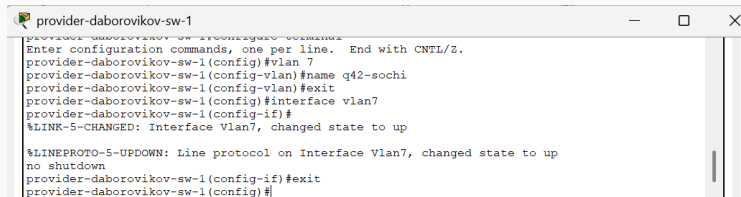
sch-sochi-daborovikov-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.128.255.5 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 15 subnets, 3 masks
O 10.128.0.0/24 [110/2] via 10.128.255.5, 00:04:28, FastEthernet0/0.6
O 10.128.1.0/24 [110/2] via 10.128.255.5, 00:04:28, FastEthernet0/0.6
O 10.128.3.0/24 [110/2] via 10.128.255.5, 00:04:28, FastEthernet0/0.6
O 10.128.4.0/24 [110/2] via 10.128.255.5, 00:04:28, FastEthernet0/0.6
O 10.128.5.0/24 [110/2] via 10.128.255.5, 00:04:28, FastEthernet0/0.6
O 10.128.6.0/24 [110/2] via 10.128.255.5, 00:04:28, FastEthernet0/0.6
O 10.128.255.0/30 [110/2] via 10.128.255.5, 00:04:28, FastEthernet0/0.6
C 10.128.255.4/30 is directly connected, FastEthernet0/0.6
O 10.128.255.6/32 is directly connected, FastEthernet0/0.6
L 10.129.0.0/24 [110/3] via 10.128.255.5, 00:04:28, FastEthernet0/0.6
O 10.129.1.0/24 [110/3] via 10.128.255.5, 00:04:28, FastEthernet0/0.6
```

Figure 8: Проверка состояния протокола OSPF на маршрутизаторе

Настройка интерфейсов

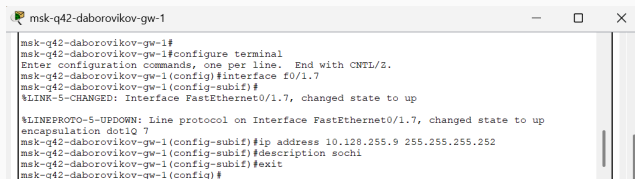


```
provider-daborovikov-sw-1
Enter configuration commands, one per line. End with CNTL/Z.
provider-daborovikov-sw-1(config)#vlan 7
provider-daborovikov-sw-1(config-vlan)#name q42-sochi
provider-daborovikov-sw-1(config-vlan)#exit
provider-daborovikov-sw-1(config)#interface vlan7
provider-daborovikov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up
no shutdown
provider-daborovikov-sw-1(config-if)#exit
provider-daborovikov-sw-1(config)#
```

Figure 9: Настройка интерфейсов коммутатора provider-daborovikov-sw-1.

Настройка маршрутизатора

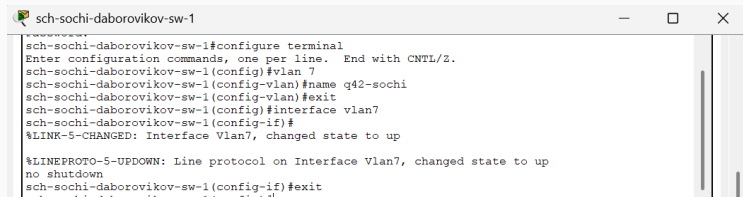


```
msk-q42-daborovikov-gw-1#
msk-q42-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-daborovikov-gw-1(config)#interface f0/1.7
msk-q42-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.7, changed state to up
encapsulation dot1Q 7
msk-q42-daborovikov-gw-1(config-subif)#ip address 10.128.255.9 255.255.255.252
msk-q42-daborovikov-gw-1(config-subif)#description sochi
msk-q42-daborovikov-gw-1(config-subif)#exit
msk-q42-daborovikov-gw-1(config)#
```

Figure 10: Настройка маршрутизатора msk-q42-daborovikov-gw-1.

Настройка коммутатора

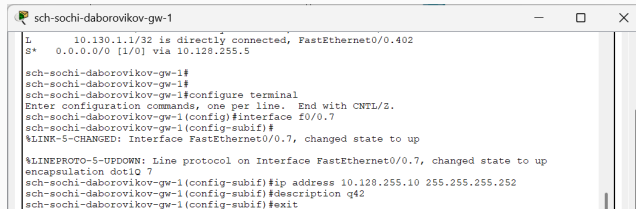


```
sch-sochi-daborovikov-sw-1
sch-sochi-daborovikov-sw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-daborovikov-sw-1(config)#vlan 7
sch-sochi-daborovikov-sw-1(config-vlan)#name q42-sochi
sch-sochi-daborovikov-sw-1(config-vlan)#exit
sch-sochi-daborovikov-sw-1(config)#interface vlan7
sch-sochi-daborovikov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up
no shutdown
sch-sochi-daborovikov-sw-1(config-if)#exit
```

Figure 11: Настройка коммутатора sch-sochi-daborovikov-sw-1.

Настройка маршрутизатора



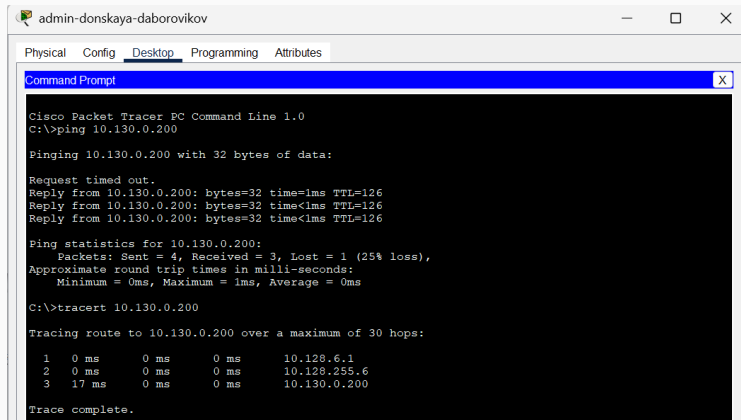
```
sch-sochi-daborovikov-gw-1
L      10.130.1.1/32 is directly connected, FastEthernet0/0.402
S*    0.0.0.0/0 [1/0] via 10.128.255.5

sch-sochi-daborovikov-gw-1#
sch-sochi-daborovikov-gw-1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-daborovikov-gw-1(config)#interface f0/0.7
sch-sochi-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.7, changed state to up
encapsulation dot1Q 7
sch-sochi-daborovikov-gw-1(config-subif)#ip address 10.128.255.10 255.255.255.252
sch-sochi-daborovikov-gw-1(config-subif)#description q42
sch-sochi-daborovikov-gw-1(config-subif)#exit
```

Figure 12: Настройка маршрутизатора sch-sochi-gw-1.

Ping



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'admin-donskaya-daborovikov'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The Command Prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.130.0.200

Pinging 10.130.0.200 with 32 bytes of data:

Request timed out.
Reply from 10.130.0.200: bytes=32 time=1ms TTL=126
Reply from 10.130.0.200: bytes=32 time<1ms TTL=126
Reply from 10.130.0.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>tracert 10.130.0.200

Tracing route to 10.130.0.200 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.128.6.1
  1  0 ms    0 ms    0 ms    10.128.255.6
  2 17 ms    0 ms    0 ms    10.130.0.200

Trace complete.
```

Figure 13: Ping по адресу 10.130.0.200.

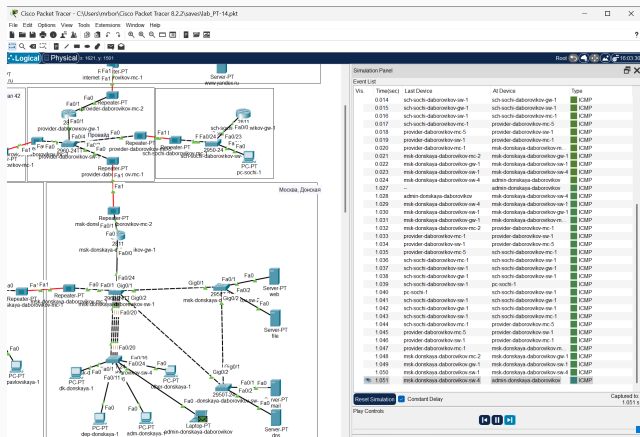


Figure 14: Отслеживание в режиме симуляции движения пакета ICMP с ноутбука администратора сети на Донской в Москве до компьютера пользователя в филиале в г. Сочи.

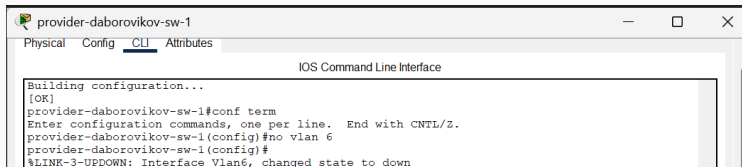


Figure 15: Временное отключение на коммутаторе провайдера vlan 6.

Проверка изменения маршрута

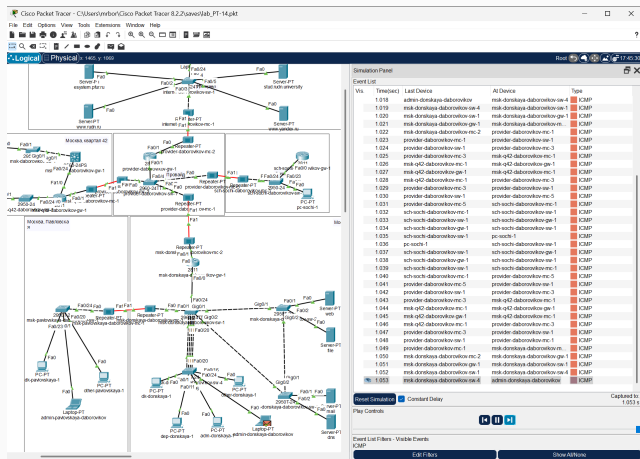
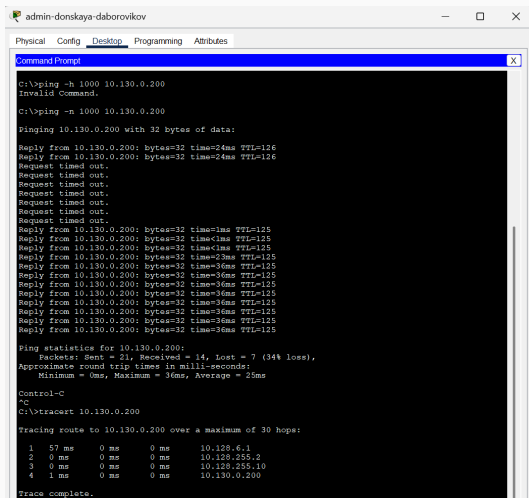


Figure 16: Проверка изменения маршрута прохождения пакета ICMP в режиме симуляции с ноутбука администратора сети на Донской в Москве до компьютера пользователя в филиале в г. Сочи.

Потеря пакетов



```
admin-donskaya-daborovikov
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping -h 1000 10.130.0.200
Invalid Command.

C:\>ping -n 1000 10.130.0.200

Pinging 10.130.0.200 with 32 bytes of data:
Reply from 10.130.0.200: bytes=32 time=24ms TTL=126
Reply from 10.130.0.200: bytes=32 time=24ms TTL=126
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Reply from 10.130.0.200: bytes=32 time=1ms TTL=125
Reply from 10.130.0.200: bytes=32 time<1ms TTL=125
Reply from 10.130.0.200: bytes=32 time<1ms TTL=125
Reply from 10.130.0.200: bytes=32 time=23ms TTL=125
Reply from 10.130.0.200: bytes=32 time=36ms TTL=125
Reply from 10.130.0.200: bytes=32 time=36ms TTL=125
Reply from 10.130.0.200: bytes=32 time=36ms TTL=125
Reply from 10.130.0.200: bytes=32 time=36ms TTL=125
Reply from 10.130.0.200: bytes=32 time=36ms TTL=125
Reply from 10.130.0.200: bytes=32 time=36ms TTL=125
Reply from 10.130.0.200: bytes=32 time=36ms TTL=125
Reply from 10.130.0.200: bytes=32 time=36ms TTL=125
Reply from 10.130.0.200: bytes=32 time=36ms TTL=125

Ping statistics for 10.130.0.200:
    Packets: Sent = 21, Received = 14, Lost = 7 (34% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 36ms, Average = 25ms

Control-C
^C
C:\>tracert 10.130.0.200

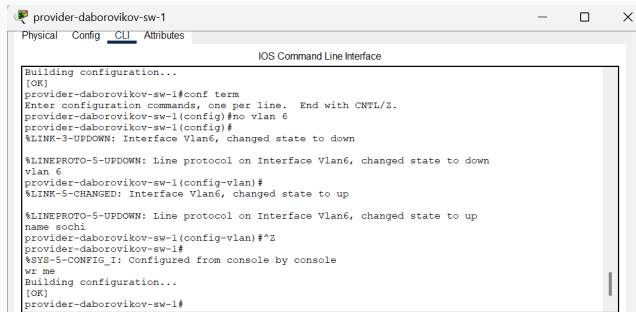
Tracing route to 10.130.0.200 over a maximum of 30 hops:

  0  57 ms  0 ms  0 ms  10.128.6.1
  1  0 ms   0 ms  0 ms  10.128.255.2
  2  0 ms   0 ms  0 ms  10.128.255.10
  3  1 ms   0 ms  0 ms  10.130.0.200

Trace complete.
```

Figure 17: Потеря пакетов.

vlan 6.



```
provider-daborovikov-sw-1
Physical Config CLI Attributes
IOS Command Line Interface
Building configuration...
[OK]
provider-daborovikov-sw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
provider-daborovikov-sw-1(config)#no vlan 6
provider-daborovikov-sw-1(config)#
%LINK-3-UPDOWN: Interface Vlan6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to down
vlan 6
provider-daborovikov-sw-1(config-vlan)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up
name sochi
provider-daborovikov-sw-1(config-vlan)#^Z
provider-daborovikov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr me
Building configuration...
[OK]
provider-daborovikov-sw-1#
```

Figure 18: Восстановление на коммутаторе провайдера vlan 6.

Проверка изменения маршрута

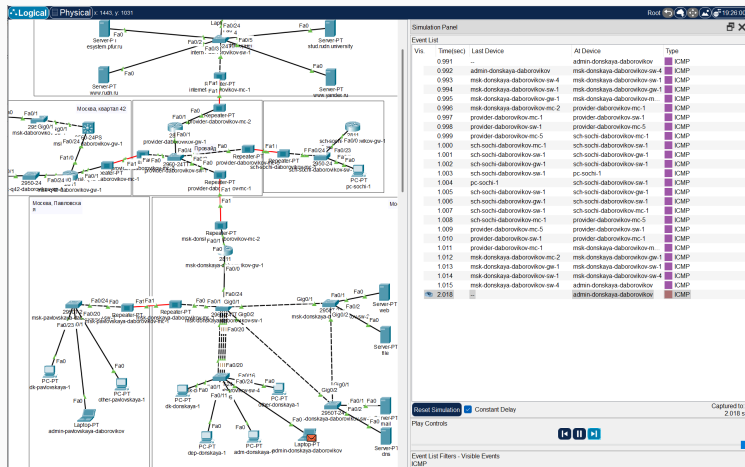


Figure 19: Проверка изменения маршрута прохождения пакета ICMP в режиме симуляции с ноутбука администратора сети на Донской в Москве до компьютера пользователя в филиале в г. Сочи.

В ходе выполнения лабораторной работы мы настроили динамическую маршрутизацию между территориями организации.